

Model Selection & Architecture Design

Date	2 October 2026
Team ID	SWTID-2026-6582
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	Project Deliverable

Selected Model

Google Gemini 2.5 Flash Lite is the configured model in the verified implementation, accessed through the official google-genai Python SDK.

Reason for Model Selection

- Supports fast generative responses suitable for interactive web use.
- Supports structured recommendation generation.
- Supports multimodal input required for outfit-image analysis.
- Integrates directly with the Python backend through the Google GenAI SDK.

Workflow Design

User → validated planner input → structured prompt (+ image when applicable) → Gemini → typed response → deterministic Python budget validation → optional budget revision → search-link enrichment → persistence → rendered result.

API Integration Architecture

The Gemini client is centralized in `app/services/gemini_service.py`. Planner routers do not call the model directly; `app/services/recommendation_service.py` orchestrates model calls, validation and retries.

Token / Context Handling

Prompts include only the planner context needed for the current request and request structured output. No long conversational memory is required for planner generation.

Libraries & Tools Used

google-genai, FastAPI, Pydantic, SQLAlchemy, Pillow, Jinja2, bcrypt, PyJWT, Uvicorn and Pytest.

Scalability Considerations

The service separation allows future migration to managed databases, cloud deployment, caching, queues and merchant APIs without changing the core planner concepts.